Manifold-Informed Architecture Benchmark

Dataset: CIFAR10 | Generated: 2026-03-29 00:35:25

Executive Summary

```
Dataset           :  CIFAR10
Input dimension   :  3,072D  (ambient pixel space)
Classes           :  10
Intrinsic dim (d) :  34  (98.9% of ambient dims are noise)
Variance threshold:  τ = 0.9
```

Training Protocol

```
Epochs            :  50
Trials            :  5
Optimizer         :  Adam
```

Best Architecture

```
PCA→34D + MLP (2d→d)
Accuracy  :  0.4899 ± 0.0034
Params    :  5,076
Eff       :  0.0965 acc / Kparam
```

vs Standard Architecture

```
Accuracy delta    :  +0.0119  (+1.19 pp)
Parameter ratio   :  161.7x fewer parameters
Efficiency gain   :  165.8x acc/Kparam
```

Provenance & Environment

Repository

```
Project  :  proteusPy

URL      :  https://github.com/suchanek/proteusPy

Commit   :  d75f66ee  (--abbrev-re
d75f66ee23710c2532ea5fe46bd3588c95e40517)
Date     :  2026-03-29 00:28:46 -0400

Message  :  add: unified test structure, outputs
```

Software

```
Python       :  3.12.13

TensorFlow   :  2.16.2

Platform     :  macOS-26.4-arm64-arm-64bit

Host         :  Turing
```

Hardware

```
Machine      :  Apple M5 Max MacBook Pro, 64 GB RAM, 2TB SSD

Compute      :  CPU
```

Manifold Discovery Parameters

```
k neighbors (local PCA)  :  ?

Discovery samples        :  ?

Samples per class        :  ?

Variance threshold τ     :  0.9
```

Results Files

```
JSON   :  cifar10_architecture_results.json

Figure :  cifar10_architecture_results.png

Report :  cifar10_report.pdf

Report :  cifar10_report.md
```

```
Report generated  :  2026-03-29 00:35:25
```

Manifold Discovery

Local PCA | k=? | τ=0.9

Intrinsic Dimensionality vs Variance Threshold

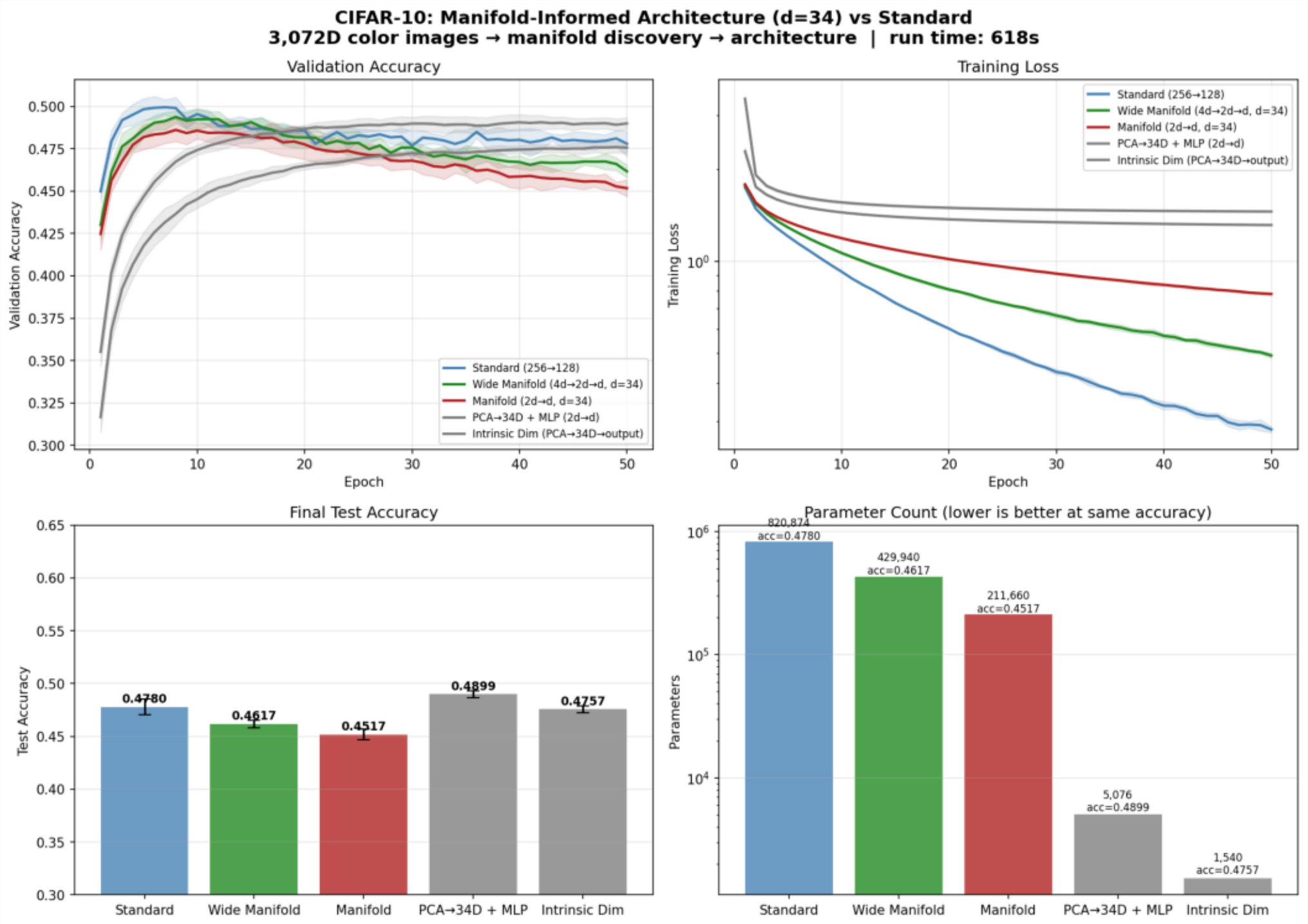
τ	Mean d	Std	Min	Max	Noise %
0.95	36.1	1.8	29	40	98.8%
0.90	29.0	1.9	22	33	99.1%
0.85	23.8	1.9	17	28	99.2%
0.80	19.8	1.8	13	24	99.4%

Selected d = 34 (max per-class max at τ=0.9, clamped to n_classes=10)

Per-Class Intrinsic Dimensionality

Class	Mean d	Std	Min	Max
frog	31.7	1.5	28	34
truck	31.6	1.3	28	34
automobile	31.6	1.4	25	33
horse	30.6	1.1	28	33
deer	29.0	1.1	26	31
bird	28.7	2.0	24	32
cat	28.5	1.0	26	30
dog	28.3	0.8	27	30
airplane	26.1	1.7	22	29
ship	25.6	2.1	21	29

CIFAR10 — Architecture Comparison Results



Architecture Comparison

CIFAR10 | 50 epochs | 5 trials | Adam optimizer

Results Table

Architecture	Params	Test Acc (mean±std)	Loss	Acc/Kparam
Standard (256→128)	820,874	0.4780 ± 0.0074	4.0688	0.0006
Wide Manifold (4d→2d→d, d=34)	429,940	0.4617 ± 0.0037	2.9960	0.0011
Manifold (2d→d, d=34)	211,660	0.4517 ± 0.0049	2.2602	0.0021
✦PCA→34D + MLP (2d→d)	5,076	0.4899 ± 0.0034	1.4557	0.0965
Intrinsic Dim (PCA→34D→output)	1,540	0.4757 ± 0.0033	1.4732	0.3089

✦ = best performing architecture

Parameter Efficiency Ranking (acc / Kparam)

1. Intrinsic Dim (PCA→34D→output) 0.3089 (0.4757 / 1,540)
2. PCA→34D + MLP (2d→d) 0.0965 (0.4899 / 5,076)
3. Manifold (2d→d, d=34) 0.0021 (0.4517 / 211,660)
4. Wide Manifold (4d→2d→d, d=34) 0.0011 (0.4617 / 429,940)
5. Standard (256→128) 0.0006 (0.4780 / 820,874)

Summary vs Standard

Best architecture : PCA→34D + MLP (2d→d)
Accuracy delta : +0.0119 (+1.19 pp)
Parameter ratio : 161.7x fewer parameters
Efficiency gain : 165.8x acc / Kparam

Key Findings & Interpretation

Manifold Hypothesis Test Results

The CIFAR10 training set occupies a 34-dimensional manifold
embedded in 3,072-dimensional pixel space.
98.9% of ambient dimensions carry no structure.

Architecture Performance

A manifold-informed architecture (PCA→34D + MLP (2d→d))
achieves 0.4899 test accuracy — 1.19 pp better
than the Standard baseline (0.4780),
using 161.7x fewer parameters (5,076 vs 820,874).

Parameter efficiency improves 165.8x over Standard.

PCA Pre-Projection Insight

When ambient noise exceeds 90%, PCA pre-projection to d=34
removes uninformative variance before the network trains.
Networks operating on raw 3,072D input must learn this
projection implicitly, wasting capacity on noise dimensions.

Interpretation

These results support the manifold hypothesis for classification:
sizing network bottlenecks to the intrinsic dimensionality of the
data manifold — rather than the ambient input dimension — yields
equal or better accuracy with substantially fewer parameters.

The parameter efficiency advantage scales with the ambient-to-
intrinsic dimensionality ratio (compression factor).